



Scratch Conditionals & Decision Trees

Expert answer key — all model responses. Accept equivalent wording.

SCRATCH

IF / IF-ELSE

AI CONNECTION

DECISION TREES

✓ ANSWER SHEET

NAME

ANSWER KEY — All Sections

DATE

Jul 6, 2026

SECTION

Section 1 / 2 / 3

📋 **INSTRUCTOR / SELF-GRADING NOTE:** Bolded text = key concept students must include for full credit. Accept equivalent wording for all open-ended responses.

🧩 PART 1 — WARM-UP · NO SCREENS

Your Brain Already Uses Conditionals

📌 FILL IN THE BLANKS

IF it is raining, THEN I will ____

bring an umbrella / wear a raincoat

(any sensible protective action)

IF I finish homework, THEN ____ ELSE ____

THEN: play / watch TV

ELSE: keep working / stay in

IF sprite touches edge, THEN ____

bounce / stop / say 'Oops!'

(any edge-response action)

💬 TURN & TALK

"A conditional is like a fork in the road because..."

✓ **"A conditional is like a fork in the road because** the program must choose one path based on whether the condition is true or false — just like you choose which road to take depending on your destination."

Award full credit if student mentions: **choosing a path + based on a condition.**

📖 PART 2 — VOCABULARY CHECK

Words We Use Today

🔑 MATCH THE TERM — ANSWER KEY

TERM	ANS	DEFINITION
1. Conditional	✓ A	A yes/no question that routes a program
2. Boolean	✓ D	A true/false value
3. Branch	✓ B	The code path taken when a condition is met
4. if-else	✓ C	An if block with a true and false path
5. Decision Tree	✓ E	A flowchart of nested decisions

🌐 CIRCLE T OR F — ANSWER KEY

STATEMENT	ANS	EXPLANATION
An if block ALWAYS runs its inside code.	○ F	Only runs when condition is TRUE.
An if-else runs ELSE when condition is FALSE.	✓ T	ELSE handles the FALSE case.
A Boolean can be "maybe" — not just true/false.	○ F	Booleans are strictly binary: TRUE or FALSE only.
An AI classifier is basically a large decision tree.	✓ T	Many classifiers chain thousands of if/else decisions.

🛠️ PART 3 — BUILD IT! GRADING GUIDE

Open Build — Conditionals in Action

☆ TIER A — STARTER

CHECKLIST — AWARD 1PT PER ITEM:

- Sprite selected & named in project.
- **when** [space] pressed → play sound block present.
- **if** block with **<key [space] pressed?>** condition visible.
- Inside **if** : **say** block with a **custom personal message** (not default 'Hello!')
- Personal message written in 'What I built' field.

✓ Full credit: all 5 items confirmed + personalized message.

☆☆ TIER B — EXPLORER

CHECKLIST — AWARD 1PT PER ITEM:

- **forever** loop with **if** **<touching edge?>** → say [Oops!].
- **if-else** upgrade: THEN says one thing, ELSE says another.
- 2nd **if** : UP arrow → move 10 steps.
- Observation written: what happens walking into wall.

✓ Full credit: observation is specific — e.g. "sprite bounced / stopped / said Oops!"

☆☆☆ TIER C — BUILDER

CHECKLIST — AWARD 1PT PER ITEM:

- Tier B fully completed first.
- Nested **if-else** inside edge block: space → "I escaped!" / "I'm stuck!"
- **<A>** **and** **** operator triggers "Double trigger!" message.
- AI connection → **condition + decision** + **multiple classifiers** **features** (2–3 sentences): **path** **test** **must mention**

✓ Full credit: AI link explicitly connects nested ifs to layered classifier decisions.

🎫 PART 4 — REFLECTION & EXIT TICKET

What Did You Learn Today?

📄 EXIT QUESTIONS — MODEL ANSWERS

1. Difference between **if** and **if-else**?

✓ An **if block** runs its code ONLY when the condition is TRUE — if false, nothing happens. An **if-else block** always runs something: THEN when true, ELSE when false — it guarantees a response in both cases.

2. Real-life if-else decision you made today:

✓ Sample: IF I was hungry at lunch, THEN I ate my food, ELSE I saved it for later. (Accept any realistic daily decision with correct IF / THEN / ELSE structure.)

3. What is a decision tree, and how does it connect to AI?

✓ A **decision tree** is a flowchart of nested yes/no questions leading to a final answer. AI uses decision trees to **classify data** — e.g., a spam filter: Is subject suspicious? → Does it contain certain words? → Spam or Not Spam.

📊 FEELING SCALE — TEACHER GUIDE

- 😞 **Lost** — needs re-teaching next session
- 😞 **Getting there** — needs guided practice
- 😊 **I get it!** — on track, proceed normally
- 😄 **Teach it!** — ready for extension/enrichment

📝 SELF-REFLECTION SCORING

"I was good at..."

Any genuine strength earns full credit.

"I still wonder..."

Any sincere conceptual question earns full credit. Deduct if left blank.